The CTAO Open Science Data Challenge

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David Green	SDC Definitions, SDC Closing Out

Abbreviations	
CTAO	Cherenkov Telescope Array Observatory
SDC	Science Data Challenge
SDC	The Open Science Data Challenge
ToO	Target of Opportunity
NFW	Navarro-Frenk-White
FoM	Figure(s) of Merit
SAT	Science Analysis Tool(s)
VHE	Very-High-Energy
ToO	Target of Opportunity
SB	Scheduling Block
OB	Observation Block
IKC	In-Kind Contribution
DL3	Data-Level 3
DL5	Data-Level 5
DL6	Data-Level 6
SC	Steering Committee
TTG	Technical Task-Force Group
PS	Project Scientist
EBL	Extragalactic Background Light
LC	Light Curve
SED	Spectral Energy Distribution
PWN(e)	Pulsar Wind Nebula(e)
SNR	Supernova Remnant
ISM	Interstellar Medium
DGE	Diffuse Gamma-ray Emission
SFR	Star Forming Region
AGN	Active Galactic Nucleus
SMBH	Super Massive Black Hole
PBH	Primoradial Black Hole
UHE	Ultra-High Energy
GRB	Gamma-ray Burst

GW	Gravitational Wave
DM	Dark Matter
dSph	Dwarf Spheroidal Galaxy
ICM	Inter-Cluster Medium
FoV	Field of View
GPS	Galactic Plane Survey
GC	Galactic Centre
LMC	Large Magellanic Cloud
ROI	Region of Interest
IRF	Instrument Response Function

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1 Introduction

1.1 Purpose

The purpose of this document is to introduce the CTAO Open Science Data Challenge (SDC) organized by Cherenkov Telescope Array Observatory (CTAO).

1.2 Scope

The scope of the document is only applicable to the CTAO Open Science Data Challenge organized by Cherenkov Telescope Array Observatory (CTAO). Any developments made for the use in planning and executing SDC may have benefits towards CTAO Science Operations.

1.3 Applicable and reference documents

1.3.1 Applicable documents

AD-1	Computing-related Requirements for the CTAO Open SDC, SDC-D10000001200350/A001
AD-2	Layouts of the CTAO Arrays, D10000001009686, B001/F, 2024

1.3.2 Reference Documents

RD-1	Top-level Data Model Specification, D10000001009642, B001/F, 2024
RD-2	SB/OB data model, CTA–SPE–COM–000000–0001-2a
RD-3	Science Analysis Use Cases, D10000001200352/A001
RD-4	User data model, D10000001200691/A001
RD-5	Science Operation Concept, CTA-PLA-PSC-000000-0001
RD-6	SDC Detailed Timeline, D10000001200353/A001
RD-7	Proposal Data Model, D10000001200690/A001

2 Phases of SDC

The CTAO SDCs are a series of source finding and source characterization data challenges on simulated CTAO data products. They will be regular issued to the community as part of the science operations preparation activities. The purpose of these challenges is fivefold:

- To make the broader scientific community aware of the scientific capabilities of CTAO.
- To allow for the broader science community to become familiarized with the CTAO data products and the Science Analysis Tools (SATs) that allow the user to analyse science ready data (i.e. DL3 as defined in the top-level data model [RD-1]) and produce the advanced science data (i.e. DL5 as defined in the top-level data model [RD-1]) produces needed for peer reviewed publications.

- To serve as test-bench for driving forward new algorithms and technologies, such as machine learning, for event reconstruction leading to improved performances as well source detection/identification in the context of source confusion. Due to the high sensitivity of CTAO, a factor of 5-7 more sources will be detected in the Galactic plane, a majority of which are likely to be extended, which increases the chances of source confusion.
- To serve as an intermediate step in the verification of software packages that will be used during the Observatory operations and foreseen to be delivered as in-kind contributions, such the SATs, CTAO scheduler, and data model/formats.
- To help future CTAO users to become familiarized to the science operations of CTAO for future observation planning.

The realism of the simulated data products will increase in accuracy, when compared to non-simulate data, with incremental SDCs. This document focuses on the CTAO Open SDC, hereafter referred to as SDC.

3 The Open Science Data Challenge

The SDC dataset consist of DL3 data products simulating the very-high-energy (VHE) sky as seen by the CTAO **Alpha Configuration** over a seven-year period, from January 1st, 2028, until December 31st, 2034. The amount of observing time covered by the simulated dataset is about 7000 hours per array (for more detailed information see 5.1). **The time will be devoted to dark sky-brightness with some additional hours devoted to light moon ToO observations, depending on the availability of the IRFs simulated.** Sub-array observations are not considered in the current SDC.

The SDC will be the first blind and open science data challenge in VHE astronomy. In general, properties of the sources, such as spectral, spatial, and temporal, will not be known to the user. Sources names will be known for deep observations on specific targets and hidden sources will be included for large surveys. It will also be the first SDC to be publicly released, thus allowing any scientist to participate.

3.1 Goals

The SDC has a few specific goals that can be categorized as scientific and technical. From a scientific point of view, SDC goals are:

- Use the results of the challenge to define the Observatory deliverable of a theoretically computed Interstellar Emission Model (or several, computed via different methods).
- Challenge the users to determine their own Diffuse Gamma-ray Emission (DGE) model based purely on gamma-ray observation simulations (software not included in the release version 1 of the SATs).
- Assess the performance of the different source detection algorithms (defining a FoM for catalogue creating).
- Foster the preparation of analysis pipelines searching for pulsed signals.
- Raise the community's awareness of high scientific impact of the Observatory through the detection of dark matter signal either through line searches of DM dense regions or extended source detection of ZFW DM profiles.

The more technical goals are mainly related to the construction project, and are defined as:

- Provide a list of science analysis use cases.
- Provide a mock catalogue of Scheduling Blocks (SBs) to the In-Kind Contribution (IKC) team developing the CTAO Scheduling software.
- Test the optimization algorithm of Long-Term observation Scheduler (LTS) against the few constraints considered in SDC.
- Verify the requirements on the SATs considered for SDC and that will be included in first official release of the CTAO SAT.
- Prepare good documentation for the SATs including analysis examples (to be used for user support).
- Test one prototype instance of the future CTAO authentication and authorisation system.
- Build a prototype of the CTAO Science Portal, the SDC portal, access the data and SAT.

3.2 Governance

The SDC is run by the Cherenkov Telescope Array Observatory (CTAO) in collaboration with the CTAO Scientific Consortium. The main committees have been established:

- The **Steering Committee (SC)** is responsible to steer the challenge, define the goals, and prepare an execution of plans. The members of the SC are represented by the main providers of the SDC, i.e. those that will commit to provide the specific deliverables.
- The **Technical Task-Force Group (TTG)** is the executive group that will ensure the challenges happens within the desired schedule established by the SC. Its specific responsibility is to prepare the challenge, ensure its smooth execution, rank the results once the challenge officially closes, assign the winners, and prepare the closing-out documents, in the three phases as described in Section 3.2.

It is steered by the SC chaired by the CTAO Deputy Project Scientist (DPS). Figure 1 - SDC Steering Committee shows the organization chart for the SDC Steering Committee. Figure 2 - Technical Task Group Organizational Chart shows a summary of the roles in the TTG.

3.2.1 Synergies with the CTAO computing department

SDC, as defined, implies strong synergies with the CTAO computing department and with the CTAO science operations team. The schedule of the SDC has been optimized to fit the needs and priorities of the CTAO construction project. As such, the technical goals can be considered as milestones of the construction project schedule.

SDC will be prepared and run in close collaboration with the CTAO computing department that keeps the final responsibility on the delivery of any software packages in any of its intermediate stages, on the definition of data model format, and the establishment of software and data product licences.

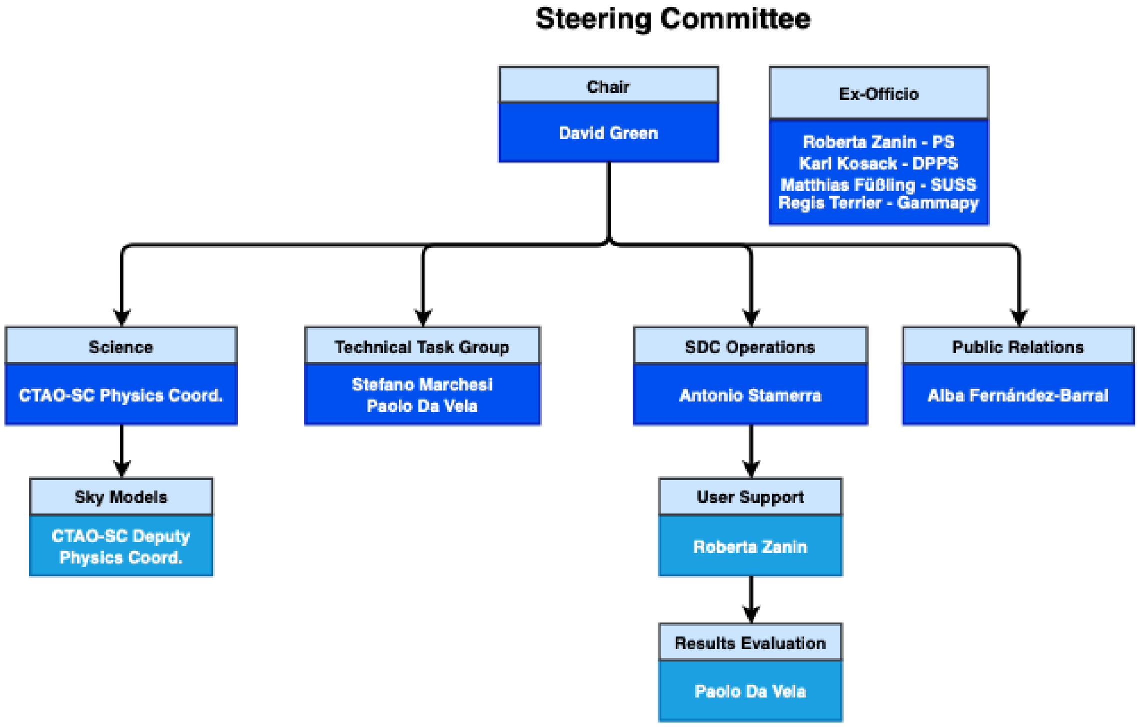

Figure 1 - SDC Steering Committee

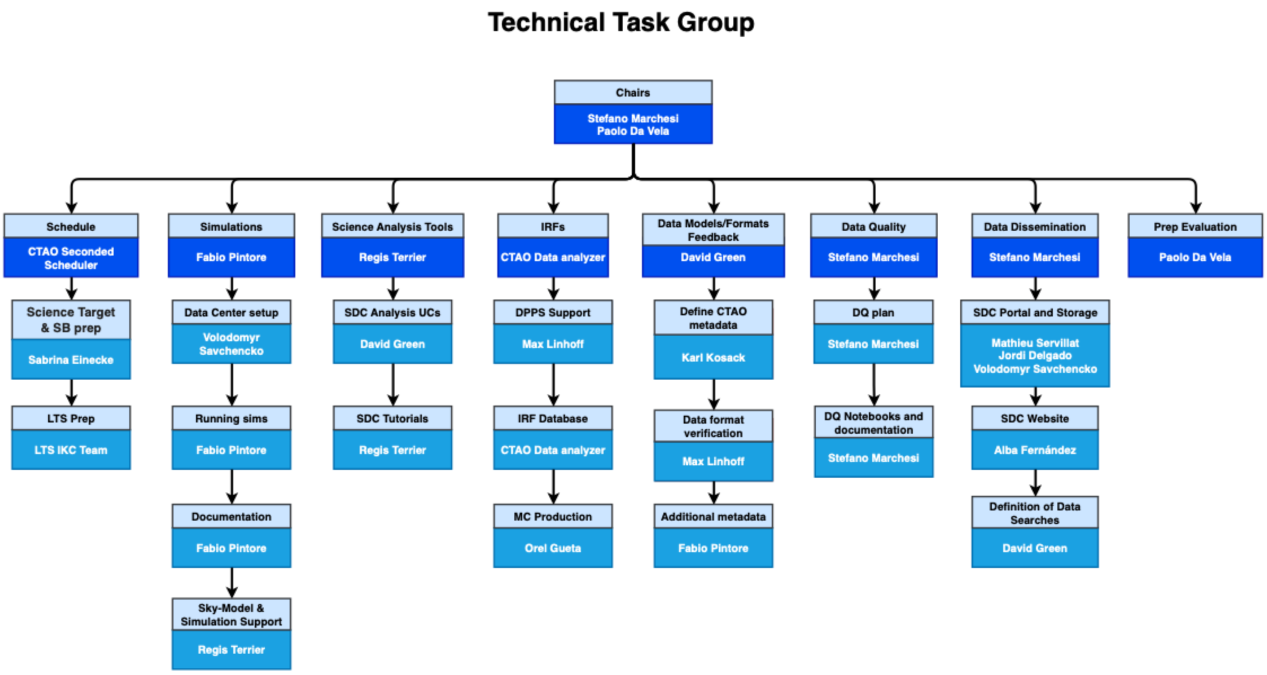

Figure 2 - Technical Task Group Organizational Chart

3.3 Pre-requisites for the SDC

The release of the SDC data implies that CTAO has already in place:

- Licence definition for the data products that is a responsibility of the CTAO Computing department
- Data protection policy that is responsibility of the CTAO Legal department

3.4 The SDC Project Schedule

The SDC is broken up into 4 phases which set the large-scale structure of the SDC.

- **Phase 1 – Definition**: consisting of the definition of the SDC goals, science cases, and technical needs
- **Phase 2 – Preparation**: consisting of the preparation and running of simulations. This step includes preparation of tools and infrastructure needed for its execution. This phase concludes when the simulated data products are officially released.
- **Phase 3 – Execution**: consisting of the release of data products to users for which they will run their analysis and submit their results. It finishes when the deadline for results submission is over.
- **Phase 4 – Closing-out**: This phase foresees the scrutiny of the submitted results, the nomination of the winners according to the a-priori defined criteria, and the writing up of a closing-out document including the lessons-learned. The preparation of a peer-reviewed publication reporting on the results of the SDC is also considered.

The timeline associated with the SDC is meant to synch CTAO software release and piggyback off the development of the said software. Final release of CTAO Data Products is scheduled for Q4 2026. A more detailed, month by month, timeline synchronized with the CTAO Computing timeline and milestones is currently in development and will be described in RD-6.

Figure 3 - Preliminary Timeline for the SDC is presents the overall timeline of the SDC and the responsible for each section. The work is split between the SDC Team, described in 3.2, Project Science, CTAO Computing, and CTAO Communications.

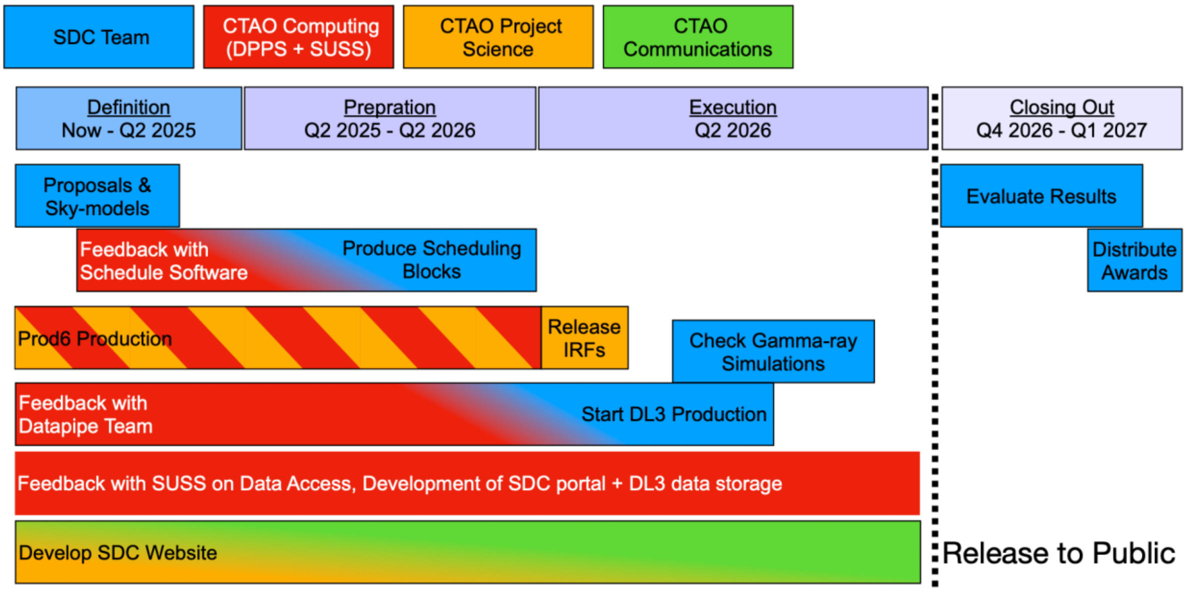

Figure 3 - Preliminary Timeline for the SDC

4 SDC Definitions

4.1 High-level Science Operation Definition and Requirements

4.1.1 Definition of Available Observing Time

In the specific case of SDC, we assume the observing period lasts 1 solar year from January 1st till December 31st. Therefore, this data challenge will cover seven observation periods. The computation of the observing time available per observing period has been obtained under the following definitions and assumptions:

- The overall astronomical nighttime is defined as the amount of time corresponding to astronomical night, the time interval between astronomical dawn and dusk. Astronomical dawn/dusk is defined as the sunrise/sunset when the geometric centre of the Sun is at an altitude angle of -18 degrees below the ideal horizon.
- The duty cycle of the Northern and Southern arrays is different among each other. Based on the average of past information about the two sites' weather conditions, the duty has been fixed to 0.9 and 0.95 for the Northern and Southern sites, respectively. The duty cycle only accounts for losses due to bad weather.
- Each site has 50 hours of observation time allocated to technical activities. These activities would make use of either the commissioning nighttime, or the engineering nighttime, or the performance monitoring time. These 50 hours are arbitrary at this early stage but have been included to more accurately represent reality. The 50 hours are subtracted from the total observation time for each period.

- Once the duty cycle and 50 hours of technical time have been taken into account, only 67% and 72% of the remaining nighttime should be considered for the Northern and Southern array, respectively, in order to approximate the observing time that is allocated to CTAO users once subtracted the host guaranteed observing time, the International Community Observing Time (ICOT), and the Directors Discretionary Time (DDT).

The SDC will include a few astronomical observations with elevated sky brightness. Since there will only be a few observations, there is no need to compute the overall CTAO nighttime, that is defined as the sum of nighttime plus the CTAO moon-time during which the CTAO arrays/subarrays can perform astronomical observations.

4.1.2 Definition of Offered Observation Capabilities

The following observation configuration setup is considered:

- **Telescope Tracking Mode:** Sideral
- **Array Pointing Mode:** Parallel
- **Observation Mode:** Wobble, On/Off, or Grid-Scan Mode
  Please note that the wobble foresees a fixed number of five wobble positions with an offset of 0.7 degrees and offset rotation angle of 0 degrees. The can mode foresees a user-defined gride of pointing positions that should by the PI of the observation projects in a machine-readable format and in the form, text describing the algorithm for determining the grid positions, if present.
- **Sub-arrays:** Northern and Southern arrays
  Only the Alpha configuration and final layout described in AD-2. Please note that if the Italian funded telescopes will be officially recognized by the CTAO ERIC Council before the start of the simulation running, we can consider, at the last minute, to use the Alpha+ which defined as the Alpha configuration plus the additional funded telescopes.

4.1.3 Scheduling Definitions

The scheduling requirements are described in AD-1. Scheduling Blocks (SB) and Observation Blocks (OB) are defined in RD-2.

In the internal SDC, a preliminary version of accepted science targets is produced by hand the SBs are created via a preliminary version of the CTAO scheduling software and the OBs are created by an ad hoc script. These are then used as input into the simulation pipeline. The script can be found at the following location:

https://gitlab.cta-observatory.org/cta-science/sdc/sdc-simulations/-/tree/main/MAKE_SDC?ref_type=heads

In the SDC, we will use a preliminary version of Accepted Science Target (RD-7) and a preliminary version of the CTAO Scheduling software to produce the SBs and OBs via the LTS and MTS. The Accepted Science Targets will come associated with scientific evaluation, score, rank, and time priority like the real CTAO proposal evaluation. ToO follow-ups of alerts will be included manually into the schedule and are not required by the MTS.

4.1.4 Scheduler Requirements

These **minimal requirements** are from AD-1 and given more context.

1. **The scheduler shall only deal with the full Northern and Southern array.**

   As described in 4.1.2, only the full Northern and Southern arrays will be used by the SDC. Further sub-array configurations are not planned on being simulated and therefore the scheduler is not required to produce a schedule for sub-array configurations but the full Northern and Southern arrays.

2. **The scheduler shall be able to schedule SB with fixed time windows to account for ToO.**

   ToOs will be included manually as their start time is not known prior to the random selection of SkyModels provided by the CTAO Consortium. After the production of the LTS by the CTAO scheduler, ToO and transient alert follow-ups time-windows and priorities will be provided to the MTS to produce the updated schedule.

3. **The scheduler shall be able to remove from the maximum available observing time to be allocated fixed time windows that simulate the loss of data because of bad weather of Host Guaranteed Time.**

   Like the 2nd requirement, the random times of bad weather and HGT will be simulated to reduce the time available to the LTS.

4. **The scheduler shall be able to allocate the available dark observing time**
5. **The scheduler shall consider as hard constraints the zenith angle range.**

   These are considered basic functionalities of the scheduler.

6. **The scheduler shall optimize the schedule on the "priority" value as well as the guaranteeing the maximum proposal completeness.**

   As described previously, proposal will be assigned a score, rank, and time priority in the format of the accepted science target. This information will be used to produce the maximum proposal completeness as required by CTAO.
   In RD-5, proposal will be evaluated and given an evaluation category.
   - CAT-1: high priority science category with all possible effort shall be made to execute associated observations. In cases where the project is partially completed within an OP, the remaining observations shall be carried over to subsequent OP.
   - CAT-2: well-ranked science targets with a best-effort approach shall be applied to these projects within the requested OP subject to resource and availability and scheduling constraints
   - CAT-3: filler science targets that maybe scheduled as filler observations depending on available resources and gaps in the schedule.

   Within each evaluation category, the science targets are given a score. A proposal is considered competed when the visibility window of its science target(s) within the relevant OP has closed and at least 95% of the requested observing time has been successfully acquired.

4.1.5 User Definitions

User in this context are defined as the Users of the SDC who will download, analyse, and submit results for evaluation. Users will need to login to the SDC portal in order both access the simulated data and submit results to be evaluated by the SDC evaluation team. This will act as a preliminary version of the CTAO Science Portal along with a preliminary version of the AAI (Authentication and Authorization Interface) to be used by CTAO Science Portal.

4.1.6 User Requirements

The user requirements are described in AD-1 and further explained within context of the Science Portal (6.3.4) and AAI (6.3.2) further in this document.

4.1.7 Data Storage Requirements

The user requirements are described in AD-1 and further explained within context of DL3 Data Storage (6.3.1) further in this document.

4.2 SDC Science Analysis Use Cases

Science Analysis Use Cases are defined in RD-2. The SDC will examine a subset of the total Science Analysis Use Cases.

4.3 SkyModels defined for Source Classes

In this chapter the science cases covered by SDC are described by the following categorization in different sources classes.

A note on high redshift sources. In general, all gamma-ray emitting sources lying at cosmological distances are affected by absorption due to the Extragalactic Background Light (EBL). Such absorption is energy-dependent and is considered in the simulation following on specific model. This model will be declared only at the end of Phase 3 to favour blindness of the challenge.

4.3.1 Gamma-ray Binaries

Gamma-ray binaries are binaries comprising of a compact companion, typically a pulsar, and a massive star. The interaction between the pulsar and the stellar winds, or decretion disk in case of the massive star being a Be type star, produces shocks where charged particles are accelerated to relativistic energies that, in turn, produce gamma-ray emission. The peculiarity of these systems is that the gamma-ray emission is dependent on the orbit of the system and therefore periodic on that timescale. The current generation of gamma-ray detectors have discovered a handful of sources belonging to this class, which are usually point-like, except for one which, having a very long orbital period of more than 50 years, produces extended emission.

Gamma-ray binaries are an excellent science case to the following CTAO capabilities (Analysis use cases):

- Production of LCs in phase bins of orbital period, i.e. phaseograms.
- Detection of phase dependent spectral variations (phase of the orbital period).
- Statistical estimates of the periodicity (Lomb-Scargle test).
- Energy dependent morphology of a micro-quasar.

4.3.2 Pulsars

Pulsars are very rapidly rotating and strongly magnetized neutron stars that emit light in form of beams, which can be observed from Earth only when crossing our line of sight. While detecting the strong and steady emission or outbursts of gamma-ray emitting sources with IACTs has become routine, pulsars are much more challenging to detect due to their weak signal, the potential dominance of foreground gamma-ray signal from the surrounding nebula, precise timing needed to fold in timing information of the pulsar typically measured via radio telescopes, and very soft spectra typically near the energy threshold of IACTs. Despite hundreds of hours of observation time, only 5 pulsars have been detected by IACTs. Recent observations of the Vela nebula have also shown the existence of a secondary component thought to be associated with Inverse Compton emission of the stripped wind from the pulsar in the 10s of TeV energy range.

Pulsars are very special science case that require specific capabilities:

- Production of phaseograms (phase of rotation period).
- Estimation of detection significance.
- Detection of phase dependent spectral variations.
- Correlations estimate between single photon/events at different energies for Lorentz invariance studies.

4.3.3 Pulsar Wind Nebulae and Supernova Remnants

Pulsar Wind Nebulae (PWNe) are powered by the rotation energy of the intrinsic pulsars that steadily lose their energy in the form of pulsar winds. The wind shocks the ambient medium and forms a PWN that emits synchrotron radiation and gamma-ray emission via Inverse Compton scatter. PWNe represent the most dominant class of gamma-ray sources. They are extended sources that can be resolved depending on their distance and age. PWNe often also exhibit energy dependent morphology as they tend of be leptonically dominated decreasing in extension and localizing closer to the pulsar at higher energies.

Supernova Remnants (SNRs) are the leftover ejecta of supernova explosions that inject both kinetic energy and gad of their destroyed progenitor star (ejecta) into its surrounding environment in the form of an expanding supersonic blast-wave. The expansion of the blast-wave and its interaction with surrounding environment and interstellar medium (ISM) can persist for tens or even hundreds of thousands of years. This phenomenon is referred to as an SNR. The blast-wave can accelerate cosmic-rays up to TeV energies thereby creating gamma-rays either through Inverse Compton processes or proton-proton interactions.

SNRs are the second source class at VHE emitter detected in our Galaxy. The gamma-ray emission from SNRs is often extended, and when resolved, shows a typical shell type morphology of the forward shock. Their spectra can vary depending on their age, ranging from 1.5 – 2.5, and often feature an exponential cutoff in at the TeV energy scale.

These two source classes provide an optimal laboratory to test the CTAO capabilities:

- Source resolution and detailed morphology studies. Different analytical models (disk, Gaussian, shell) will be considered but also for a few cases of more complex morphology taken from real data at lower wavelengths such as radio or current generation of gamma-ray detectors.
- Energy-dependent morphology.
- Energy spectra studies with measurements of spectral changes at different energies and with different spectral shapes (justified by different emission mechanisms).
- Spatial dependent spectroscopy.
- Systematic approach in the source description to be able to perform population studies.

These studies will provide powerful insights into the mechanisms at the base of VHE emission allowing for the disentangling between leptonic and hadronic origin of gamma rays.

4.3.4 Diffuse Gamma-ray Emission

Diffuse Gamma-ray Emission (DGE) is the largest emission component at high energies, in the 100 MeV – 100 GeV sky and it has several main components. 1) the unresolved astrophysical sources, 2) the gamma-ray emission produced by the interaction of cosmic rays with ambient has and radiation fields. Gamma rays can, therefore, be considered as tracers of the bulk cosmic-ray population.

The energy spectrum of this component is soft with a flux decreasing at increasing energies. Despite the non-dominance of this component at VHE, it is essential to derive it correctly to hand in a proper way the source confusion and ability to correctly determine the spectral and morphological characteristics of foreground astrophysical sources. In additional, CTAO will provide a measure across the Galactic plane of diffuse gamma-ray emission at TeV energy scales with unrivalled angular resolution.

The second component, dubbed as Interstellar Emission Model (IEM) can be estimated by using cosmic-ray propagation code such as GalProp, DRAGON-2, or PICARD. The Observatory will release few IEMs together with SATS to allow users to run their analyses and estimate the correct level of IEM and therefore as well as the impact of unresolved astrophysical sources on the DGE.

The science case will allow the preparation of a full set of IEMs which be used to estimate the systematic error due to the uncertainty on the IEM itself. Specifically, this will allow to test the CTAO capabilities:

- Estimation of flux level of the DGE per energy and spatial bin.
- Estimation of the percentage of emission due to unresolved sources on the DGE.
- Test the bracketing systematic error computation due the uncertainty on the IEM.

4.3.5 Star-forming Regions

Although the usual assumption is that supernovae are the dominant source of cosmic rays in the Galaxy, their inability to reach the desired cosmic-ray energy of several PeV has contributed to several competitors. One such are star-forming regions (SFRs). Stellar winds from massive young stars in star clusters imbedded with SFRs have the capability of accelerating cosmic-rays to almost PeV energies and have been demonstrated as gamma-ray emitters. However, the lack of detection sensitivity for all but the largest star forming regions prevents the detailed study and clear identification of these sources in VHE energy range.

SFRs within our Galaxy are the optimal science case for very extended (more than one degree) and bright gamma-ray emitting sources. Specifically, this will allow to test the CTAO capabilities:

- Detailed morphological studies.
- Energy spectral studies with measurements of spectral changes at different energies and with different spectral shapes (justified by different emission mechanisms).
- Spatial-dependent spectroscopy.

4.3.6 AGN Population Study

One of the major unanswered questions of modern AGN physics is existence or non-existence of the blazar sequence. The blazar sequence is the phenomenological evidence of a negative correlation between the peak of the synchrotron emission frequency and the overall luminosity of the blazar. Several arguments have been made that the blazar sequence is purely a result of observation biases, but it remains widely used within the AGN community. If the sequence exists, there should be no, or only a few, detection of high-luminosity blazars that have their synchrotron peak in the optical and UV wavebands. A population study at VHE will be able to probe this prediction. In addition, simultaneous high-precision measurements of both unabsorbed intrinsic (GeV) and attenuated (TeV) sections of the blazar spectra, may allow to disentangle intrinsic physical processes from external absorption features, which will be essential for precision EBL studies.

An AGN population study would be able to test CTAO capabilities of:

- Energy spectra studies with measurements of spectral changes at different energies and with different spectral shapes (justified by different emission mechanisms).
- Very soft spectral measurements for high redshift-blazars.
- Testing different EBL models within SATs.

4.3.7 Longer-term variability of AGN

VHE observations of AGN harbouring Super Massive Black Holes (SMBHs) and ejecting relativistic outflows represent a unique tool to probe the physics of extreme environments including accretion physics, jet formation, relativistic interaction processes, and general relativity. AGNs are known to emit variable radiation across the entire electromagnetic spectrum up to multi-TeV energies, with fluctuations on timescales of several years down to several minutes. A better understanding of the emission region is directly linked to information about the possible cosmic-ray acceleration and gamma-ray emission mechanisms. Sampling years long un-biased AGN LCs and comparing to LCs at other wavelengths is crucial for distinguishing between these different emission components.

The monitoring program of AGN allows for the investigation of the long-term variability of these sources, in turn to test the following CTAO capabilities:

- Detection of time-dependent signal detection.
- Detection of time-dependent spectral variations.
- Production of long-term LCs, integral or energy flux as a function of time for different time bins and energies ranges.
- Use of more complex temporal binning algorithms within SAT such Bayesian-Blocks.
- Detection of weak extended emission around the central source.

4.3.8 Short-term variability of AGN

Investigating AGN variability on short time scales allow us to constrain the bulk Doppler factor, as well as to investigate the emission mechanisms, and particle acceleration and cooling processes. Very rapid variability, on the order of minutes, has so far been studied in a few extreme flares, but these works showed the limitations of standard emission models and gave rise of more sophisticated, e.g. multi-zone emission or magnetic reconnection models. In addition, the faster the variability, the smaller the emission region and with multi-wavelength datasets can help further constrain emission processes. Blazar flares can also be used to search for Lorentz invariance violations, as well as cast light on the physics of the ultra-relativistic inner jets of these systems. The science case includes the follow-up of ultra-high-energy (UHE) neutrino events, which are expected be correlated with flaring blazars.

Target of Opportunity (ToO) observation as following-up alerts of AGN detected a high-flux state at different wavelengths, such as X-ray or optical, allow to test the following CTAO capabilities:

- Detection of signal in response to certain follow-up and expressed as a function of time.
- Detailed study of the energy spectra and detection of possible spectral features.
- Production of LCs, i.e. integral or energy flux as a function of time for different temporal or energy bins.
- Detection of the shortest statistically significant flux and/or spectral variation.
- Computation of integral differential flux upper limits for different temporal bins.

4.3.9 Gamma-ray Bursts

Gamma-ray bursts (GRBs) are the most powerful explosions in the Universe and are by far the most luminous sources of light known since the Big Bang. The recent detection of GRB emission in the VHE range opened a new window of the investigation of particle acceleration and radiation properties in the most energetic domain. Know to occur at cosmological distances, they can also serve as important probes of the EBL, intergalactic magnetic fields (IGMF), and the fundamental nature of space-time. The potential discover space includes detection of GRBs from their onset and consequently improved tests of Lorentz invariance violation.

From GRBs and for the perspective of CTAO capability testing includes:

- Detection of signal to a certain follow-up, also expressed as a function of time.
- Detailed study of the energy spectra and detection of possible spectral features.
- Production of LCs, i.e. integral and energy flux as a function of time, for different time bins and energy ranges.
- Detection of the shortest statistically significant flux or spectral variations.
- Computation of integral and differential flux upper limits.

4.3.10 Gravitation Wave Counterparts

The multi-messenger discovery of the binary neutron stare merger GW170817 demonstrated the promise of follow-up efforts to identify and interpret counterparts of Gravitational Wave (GW) sources. In addition, the recent discovery of TeV emission from GRBs confirmed that emission form these transients can extend to the VHE range. GW event follow-up observations by CTAO may be able to probe particle acceleration and high-energy emission from binary black hole and neutron star mergers, although the former is not expected to have significant EM emission.

From the perspective of CTAO capability testing, this science case is identical to that of Gamma-ray Bursts. The main difference between a GW follow-up and GRB follow-up is the size of the localization region. GRBs have localization region either on the size of the CTAO FoV or smaller, while GW localization region can be up to 100 square degrees.

Due to this, in addition the science case of Gamma-ray Bursts, GW will also test CTAO capabilities:

- Detection of sources in a tiled pointing strategy covering a localization region larger than that of the CTAO FoV

4.3.11 Dark Matter

The nature of Dark Matter (DM) is one of the largest outstanding questions in science. The material is known to exist due to its gravitational effects and in 5x quantity to normal baryonic matter, but very little is known of the particle natures of DM. Many hypotheses exist for dark matter; one of the most debated considers a new weakly interacting massive particle (WIMP) that would be found outside the Standard Model of Particle Physics. However, valid candidates can be found at almost any mass scale. Some of the most promising theories predict that WIMPs can annihilate or decay to produce leptons, bosons, and quarks which in turn may produce gamma rays.

The Milky Way centre is expected to be the brightest source of DM in the gamma-ray sky; however, the interplay between DM and the local baryonic content via galactic evolution makes the DM brightness uncertain. In addition, the large and bright foreground gamma-ray emission from known VHE emitters and DGE make the detection of DM within the Galactic Centre very challenging. Despite these challenges, line searches can be performed on the Galactic Centre data which have reduced dependencies on our understanding of intrinsic astrophysics of the Galactic Centre.

On the other hand, Dwarf Spheroidal Galaxies (dSph) should provide a near background free signal at the expense of reduced DM densities with respect the Galactic Centre. dSphs require detailed estimates of the mass measurements, the so-called J-factor, to properly estimate the contribution of DM vs baryonic matter. J-factors can only be measured via long term monitoring of stars orbiting the candidate dSph. These are provided by large collaborations such the Dark Energy Survey (DES).

Finally, another class of targets are Milky Way DM sub-halos, which are DM clumps not associated to visible targets. Such objects can be found as unidentified gamma-ray emitters as well as in serendipity searches.

This science case allows to define the CTAO capabilities in terms of DM detection and identification. From a high-level analysis point of view, this includes:

- Detection of weak DM extended halos with or without significant gamma-ray foreground.
- Detection specific spectral features that can be considered as smoking guns for DM signal, such as cutoffs or line-like features.
- Systematic approach in the detection of the same spectral feature in several sources.
- Identification of DM microscopic nature of spectral models.
- Computation of flux upper limits expressed in terms of the parameters of interest for a specific DM candidate and interaction model.

4.3.12 Galaxy Clusters

Galaxy clusters are expected to be reservoirs of cosmic ray accelerated by structure formation processes, galaxies, and AGN. The detection of diffuse synchrotron radio emission in several clusters confirms the presence of non-thermal electrons and magnetic fields permeating the Inter-Cluster Medium (ICM). While there has been no direct evidence for proton acceleration, gamma rays can prove Galaxy clusters as cosmic-ray accelerators if VHE gamma-ray emission is detected and properly modelled via neutral pion decay. Additionally, 80% of the mass of clusters is in the form of DM and therefore are considered another target for DM searches.

Galaxy clusters are the optimal science case for very extended (more than 1 degree) and weak gamma-ray emitting sources and show how the capabilities related to extended emission detection are very different depending on the flux level of the source. The analysis constrains are therefore very similar to that of Star-forming Regions.

- Detection of source extension and estimation of preferred spatial model.
- Energy-dependent morphological studies.
- Energy spectra studies with measurements of spectral changes at different energies and with different spectral functions.

4.3.13 Primordial Blackholes

Primordial black holes (PBH) are a hypothetical type of black hole born at the early stages of the Universe, fractions of a second after the Big Bang. Depending on when they are formed, PBHs could have masses as low as 10^{-5} g to 10^{5} solar masses. They have been suggested as a DM candidate. All black holes are theorized to emit Hawking radiation at rate inversely proportional to their mass. Due to their low mass, PBHs may experience run-away evaporation, creating a burst of radiation at the final phase which can be detected in the VHE range.

Due the expected fast time scales of PBH evaporation and expected very hard emission spectrum, there is no expected external trigger for a PBH, and any discovery would be entirely serendipitous. PBHs will test the capabilities of CTAO with:

- Detection of rare event without any external trigger anywhere in the CTAO FoV.
- Detection of source on short times scales, like that of a GRB.
- Detailed study of the energy spectra and detection of possible spectral features.

4.3.14 EBL measurement

Stacking analysis of AGN and blazars

4.4 Observational Projects

The simulated observations covering all targets described in the SkyModels are group in the form observational projects. The SDC has the following seven observational projects.

- **Galactic Plane Survey (GPS):** $60° \leq |l| \leq 180°$withing a latitude range of $|b| \leq 5°$
- **Galactic Centre (GC) Survey:** the inner region of the Galactic plane ($|l| \leq 60°$) with a latitude range extending up to $|b| \leq 15°$
- **Extragalactic Survey:** $|\boldsymbol{l}| \leq \mathbf{90}°$ and $b \geq 15°$
- **Large Magellanic Cloud (LMC) Survey**
- **Deep Observations of Extended Galactic Regions:** lying within the Region of Interest (ROI) of the GPS
- **Long-term monitoring of a few bright AGNs**: independent from the above-mentioned survey
- **Follow-up of AGN flares, GW, and GRB alerts**: independent from the above-mentioned survey

Each observational project includes all observational and analysis related configuration that are necessary to produce a list of SBs, and in turn one OB that will be used to produce the simulated data products. A template is included for all configuration options available in SDC has been prepared and should fill per observational project. A contact person for each observational project will be identified, and they may or may not be the contact person for the SkyModels.

5 SDC Preparation

5.1 Schedule

The SDC available observing time represents the time that will be simulated. SDC covers the first 7 years of CTAO.

5.1.1 Dark Time

For simplicity, the first version of the schedule and SBs will only consider dark time.

Dark time is defined as: a) the Sun is 18 degrees below the horizon (Sun alt < $-18°$); b) the Moon is below the horizon (Moon alt < $0°$).

Using `astropy`, the following amount of dark was calculated: a) CTAO-North: 1653 hours/year; b) CTAO-South: 1689 hours/year. For the SDC, this would mean covering the total period of 7 years; CTAO-North: 11,557 hours and CTAO-South: 11,826 hours.

5.1.2 Maximum SDC Available Observing Time

Only a fraction of the dark time computed in the previous section will be available to the SDC. To obtain the maximum observation time, the following corrections were applied.

- A duty cycle of 90% (CTAO-North) and 95% (CTAO-South) is applied.
- 50 hours of technical time is subtracted in both sites.
- The time allocated for Host Guaranteed Time, International Community Time, International Scientific Time, and Director's Discretional Time is subtracted, which is 67% (CTAO-North) and 72% (CTAO-South) of the remaining time.

Therefore, the remaining maximum observing times that are available to the SDC are: 963 hours/year for CTAO-North and 1121 hours/year for CTAO-South. For the SDC, this would be 6742 hours for CTAO-North and 7837 hours for CTAO-South.

5.1.3 Observing Time to be Simulated

As the requested observation time is smaller than the maximum SDC available observation time, and as different iterations of the schedule may or may not include transient sources, only a fraction of the maximum SDC available observation time will be simulated and correspondingly be considered into the Scheduler.

5.2 Instrument Response Functions

To generate the gamma-ray simulated sky as seen by CTAO, one needs to convolve the SkyModels with the Instrument Response Functions (IRFs) of CTAO-North and CTAO-South. For the SDC we will be using the publicly released IRFs that will result from the ongoing Monte Carlo prod-v6. This production will have:

- Three zenith distance:, $20°$, $40°$, $60°$
- Two azimuthal angles: $0°$ and $180°$
- Two sky-brightness bins corresponding to dark and half-moon conditions

The format of the IRFs will be spatially dependent DL3 data format.

5.2.1 Event-types

Ideally, the use event types will be used in the SDC but is limited to its availability to be produced by the CTAO software. If IRFs for the event types cannot be produced within the desired SDC timeframe, then they will be included in future SDCs.

6 SDC Execution

6.1 Source Simulation

The simulated data products include all the science cases described in Section 4.3 in the form of SkyModels.

6.1.1 SkyModels

SkyModels include a collection of spectral and spatial characteristics of each considered source in the sky. This information is released in machine-readable format to be easily accessible by a wide audience and usable by a board range of software packages. At the time of writing, the official CTAO DL6 data format has not been released. Therefore, we adopt the format defined by *gammapy* and collect information by using the JSON representation. All received SkyModels will be collected in the *gammapy* format in the SDC repository, but they will be publicly released only at the end of phase 3. A copy of these models is also stored in the data centre that will be used to run the simulations.

SkyModels will be prepared by the Science Working Groups of the CTAO Scientific Collaboration (CTAO-SC) and will be described in detail in an additional document, currently in preparation, that will be kept private during the duration of the SDC. To guarantee the blindless of the challenge to all CTAO-SDC members, one contact person per SkyModel will be identified and committed to confidentiality: they will deliver the model as a large set of model instances among which the TTG would select from, or as a python code that will be run by the TTG to generate instances of the model.

It will be the responsibility of the TTG to convert the SkyModels into the *gammapy* format. The TTG commits itself to convert these models from the *gammapy* format to the final CTAO DL6 data format before public release at the end of phase 3. When publicly released, the SkyModels will be published in zenodo in a similar manner to CTAO IRFs. The license to be included in the zenodo repository will be defined by the CTAO Computing department.

The dataset is simulated using Look-Up Tables (LUTs) dealing with the main observational parameters:

- **Zenith distance: θ**
  The following three bins are considered: $\theta \geq 35°$, $50° \geq \theta \geq 35°$, and $65° \geq \theta \geq 50°$. The IRFs for each θ bin is computed by finding the average value between the IRF production zenith pointings of $20°$,$40°$, and $60°$.
- **Azimuthal angle: ψ**
  For simplicity, the IRFs will use the average azimuthal IRFs
- **Sky-brightness**
  Sky-brightness bins will be considered corresponding to dark-sky and half-moon conditions. The only science cases that will use the half-moon conditions are a few **Follow-ups of AGN flares, GW, and GRB alerts** observations. The remaining observation time will use dark-sky NSB IRFs.

6.1.2 Limitations of the simulated data products

While significant effort has been expended to make realistic data produces for the SDC analysis, there are many limitations to the degree of realism that could be achieved. Some of the most apparent discrepancies are outlined below:

- The simulated dataset assumes brand-new array elements. Therefore, the systematic errors due to the aging of the telescopes are not considered.
- The simulated dataset assumes excellent weather conditions. Therefore, systematic errors due to the changing atmosphere are not considered.
- No observations with different sub-arrays in parallel at the same site is considered.
- The zenith distance and azimuthal binning used for the IRF production is rather coarse and this will cause deviations from reality especially for zenith distances larger than $40°$.

The IRFs should include interpolation along the zenith axis as much as possible, but considering the coarse zenith bins, this will be a minor effect. Future IRFs will have a much finer zenith and azimuthal binning and will allow for improved accuracy.

6.1.3 IRF Noise

The SDC will include IRF noise on individual OBs. The IRF noise is defined as a systematic shift in the affective area IRF used to produce the gamma-ray simulation but not represented within the included IRF of the DL3 dataset. The IRF on the affective area is set as a gaussian distribution with a width of ±5%. The has the effect of adding systematic uncertainty to the IRF at the level of the expected systematic uncertainties. This also adds small level of blindness to the result, as the IRF noise level will not be known to the SDC.

Future SDCs will include additional IRF noise on other IRFs than the effective area, such as the PSF, EDISP, or BKG.

6.2 Data Validation

The data validation will be performed with a series of *gammapy* notebooks which will be provided to the community after the end of SDC. These notebooks were developed during the internal SDC and can easily be adapted to the SDC.

6.3 Data Dissemination

6.3.1 DL3 Data Storage

The simulated gamma-ray data will be stored as DL3 data format with the DL3 metadata used for data queries in the SDC portal. Several TB are required to store the data and accessible by several users accessing and downloading sub-sets of the data simultaneously.

The data storage will be provided through a dCache based system and developed by SUSS.

Data storage **requirements** are described in AD-1. The specific requirements are:

1. Data storage shall have at least 2 TB.
2. The data storage shall handle at least 50000 files.

The following **requirement** in AD-1 is no longer considered valid.

1. The registered users shall have a user storage area.

6.3.2 Authentication and Authorization

A prototype version of the AAI system will be utilized for the SDC Portal. It will allow users to register with their user information and gain access to the SDC Portal. The registered information will use the user data model described in RD-4.The AAI prototype will be provided through by the AAI and SUSS.

AAI requirements are described in AD-1. The specific requirements are:

1. Only registered users shall be able to access the SDC portal.
2. The user registration shall include name, surname of the user, affiliation of the user's research institution, and an email address.
3. The login process shall have a human check to avoid fake registration.
4. The registration shall remain open for the entire execution phase of the SDC.
5. Each registered user shall have equivalent rights on all data.
6. The maximum number of users shall be 1000.
7. The user account shall be preserved until the closing-out phase of the SDC is over.

6.3.3 SDC Website

The SDC website will communicate to the broader scientific community the data availability and the sources classes simulated within the SDC. It will also communicate the parameters of the SDC established in Section 4.1. The SDC website will link towards in the SDC portal, accessible via the AAI system, where the users can access data, SAT, and additional information defined in Section 6.3.4. The SDC website keep period SDCs for users to access after the closure of this SDC.

No specific source information will be communicated in the SDC website to ensure blindness of the SDC. The SDC website will be designed by CTAO Communications with information provided by CTAO Project Science. Finally, the SDC website will include any legal information on the ownership of the datasets and the models used to provide the datasets and any licenses used to distribute the data after the completion of the SDC.

Requirements for the SDC website are described in AD-1. The specific requirements are:

1. The SDC portal shall host the CTAO logo
2. The SDC portal shall fulfil the guidelines of the CTAO visual identity

6.3.4 SDC Portal

The SDC Portal will be provided by SUSS. The SDC Portal is the only way for users to access the SDC data. It requires them to login via AAI (6.3.2) then can access the SDC Portal. The SDC Portal is a preliminary version of the CTAO Science Portal. It offers basic data searches, description of available information provided to the user, and a space to submit results. The SDC Portal also gives the user access to SDC tutorials, SAT, and IRFs.

The Science Portal will provide a list Accepted Science Targets, their associated proposal IDs, contact person for the proposal, and pointing positions in units of degrees.

Additionally, information on the ToOs will be provided including T0 of the alert, type of alert, associated proposal, and pointing positions.

The basic data search includes the following:

- **Cone search** – either through a SIMBAD/NED linked sources name or a provided source location in celestial position. The frames included are ICRS and Galactic frames. The users can also provide radius for the cone search with the default being 10 degrees.
- **Proposal ID** – the user can access the entire data set for a given proposal ID which is provided in the SDC portal. The proposal ID can include the specific operation period or the Unique Proposal ID to access all available data.
- **Time Range** – the user can provide a time range. The time range can for the search can be provided in DD/MM/YY and HH/MM/SEC or MDJ. The total valid time-range is 00:00:00 UTC January 1st, 2028, until 00:00:00 December 31st, 2034. The default being the total valid time-range.
- **Moon Level** – The choices are None, Dark, Moderate. The default being Dark.
- **CTAO Array Site –** The choices are Both, CTAO-North, CTAO-South. The default being Both.

These search terms are considered inclusive, not exclusive, the user can select any of these search terms to access the data they wish.

The requirements for the SDC Portal are defined in AD-1. The **minimal requirements** related to the SDC Portal are:

2. The registered users shall be able to search for the data by providing the coordinates of a sky position in celestial coordinate system and the radius of the region of interest in degrees.
3. The registered users shall be able to download the selected data.
4. The registered users shall be able to download the SAT.
5. The registered users shall be able to select and download the IRFs, unless they are included in the DL3 as foreseen in our data model (point to be discussed).
6. The registered users shall be able to access tutorials and/or documentation for the data analysis.

The **optional requirements**, defined in AD-1, for the SDC Portal are:

1. The registered user shall be able to select a celestial position based on the target name as defined in the SIMBAD catalogue.
2. The registered user shall be able to select data belonging to the same observational project (using the proposal ID bit).
3. The registered user shall be able to select data within a certain time window in the following format DD/MM/YY and HH/MM/SEC.
4. The registered user shall be able to select data within a certain zenith distance range.
5. The registered user shall be able to select data combining the above-described queries.
6. The registered user shall be able to select data depending on the proposal attribute ”ToO” (yes/no)
7. There shall be different user status: individual and groups.

8. The registered users shall have the possibility to select different IRFs based on predefined events types/classes.

The following **minimal requirements** are considered outside the scope of the SDC and are no longer valid.

1. There shall be different user status: individual and groups.
2. The registered users shall be able to perform interactive analysis without downloading data.
3. The registered users shall be able to perform analysis in batch mode without downloading data.

6.3.5 SDC Tutorials

A set of high-level tutorials will be developed describing the following tasks for SDC:

1. Accessing SDC data
2. Submitting SDC results
3. Loading SDC data
4. Point-like source analysis
5. Extended source analysis
6. Phase dependent analysis
7. Time-dependent analysis

These tutorials will be mainly developed by the SDC team with help from the gammapy/SAT developers. They will be hosted on a specific page on the SDC Portal and accessible to registered. After the closure the of the SDC, the tutorials will be given to the gammapy developers to use as they wish.

6.3.6 SDC Helpdesk

A simple helpdesk will be established and hosted on the SDC website. This will provide contact information for the SDC, a FAQ, access to and a discussion area where users can ask questions to experts or fellow users of the SDC. The SDC team will staff the help desk and act as experts in analysis and if any questions appear regarding the SkyModels. Finally, a ticketing system will be established where users can submit tickets on specific issues they are encountering with the SDC. The ticket system will be setup in such a way that maintains blindless of the SDC.

In the future, this help desk may act as a preliminary version of the SUSS's helpdesk, but this is not foreseen in the current SDC.

7 SDC Closing Out

7.1 Goals and Awards

The SDC will set out a specific set of Goals and Awards available to the broader Scientific community. The Goals are a basic set of Science Analysis Use Cases without any specific source target described for the Science Analysis Use Case to ensure blindness of the SDC.

The awards will be established soon.

7.2 Results Submission

To reduce overhead on security concerns with users uploading data to the SDC Portal, a dedicated zenodo community will be used as the space where Users can upload their results. Zenodo additionally offers the user to update their results with different versions. The users will then upload their zenodo DOI to submit their results for evaluation as described in Section 7.3. The latest version available before the SDC deadline will be used for results evaluation.

The following **minimal requirement** is related to results submission:

1. The SDC Portal shall have a space wherein users can submit their zenodo link for evaluation of their results.

From AD-1, the following **minimal requirements** are no longer considered valid.

1. The users shall be able to submit the obtained results to the SDC ops team to allow for a possible evaluation. The material to be submitted can be either DL5/DL6 data products, jupyter notebooks or both.
2. The SDC portal shall display the date of the data release, and of the SDC closure, as well as the instructions on how to submit the obtained results.

7.3 Results Evaluations

Results submitted to the SDC will be evaluated by the SDC team. They will compare results from individual sources against their simulated model through a set of automated scripts and notebooks. For population studies of surveys described in Section 4.4, the SDC team will use the parameters of the source population provided to them from the model makers to compare against the submitted results.

The set of specific goals provided to the community will be evaluated according to accuracy to the source model, wherein most accurate result will win the awards. If multiple submissions have similar accuracy, then results will be ranked on earliest submitted results.

After the closure of the SDC and given awards, the source models used in the SDC will be made public for users to check their results against the simulated data.

7.4 Closing-out Document

A closing out document will be made after the conclusion of the SDC. This closing out document will be drafted by the SDC Steering Committee. It will include the number of participates, summary of a SDC short survey given to everyone who submits results, and summary of improvements for future SDCs.